

Hydrolysis and Solvolysis as Benign Routes for the End-of-Life Management of Thermoset Polymer Waste

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Abstract

The production of thermoset polymers is increasing globally owing to their advantageous properties, particularly when applied as composite materials. Though these materials are traditionally used in more durable, longer-lasting applications, ultimately, they become waste at the end of their usable lifetimes. Current recycling practices are not applicable to traditional thermoset waste, owing to their network structures and lack of processability. Recently, researchers have been developing thermoset polymers with the right functionalities to be chemically degraded under relatively benign conditions postuse, providing a route to future management of thermoset waste. This review presents thermosets containing hydrolytically or solvolytically cleavable bonds, such as esters and acetals. Hydrolysis and solvolysis mechanisms are discussed, and various factors that influence the degradation rates are examined. Degradable thermosets with impressive mechanical, thermal, and adhesion behavior are discussed, illustrating that the design of material end-of-life need not limit material performance.

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1. INTRODUCTION

Thermoset polymers, highly crosslinked and glassy polymers comprising approximately 15% of polymers produced in the United States, are increasing in worldwide production each year (1). Though thermoset polymers are typically used in durable, longer-lifetime applications, such as in composites used in aerospace and automotive industries, wind turbine blades, and structural components, their highly crosslinked network structures prevent recycling of these materials (2). As a result, most thermoset polymers are landfilled or incinerated after their useful lifetimes (3). Their growing use, and related generation of waste, may present significant environmental challenges in the future. To address these concerns, significant efforts have focused on methods of degrading thermoset polymers. The most widely studied method is pyrolysis, that is, breaking down the thermoset network in the presence of heat, sometimes promoted by catalysts and even solvents, and this approach has been extensively discussed in a recent review (4). Though most thermosets can effectively be thermally degraded (5–9), these processes are typically very energy intensive and costly, often requiring temperatures over 400°C. Photodegradation presents an alternative approach, but degradation kinetics are too slow to be of practical use for managing thermoset waste (10, 11). Innovations in the development of recyclable thermosets, such as vitrimers or other thermosets that have appropriate chemical transformations allowing for melt processing at elevated temperatures, have garnered much recent interest, and these materials have enhanced end-of-life options (12–20). A recent perspective addresses these materials, and they are not reviewed here (19).

The development of thermoset polymers that undergo hydrolysis and solvolysis under mild conditions (21–23) provides a potentially benign and low-cost route to transforming thermoset waste to useful small-molecule by-products, which could subsequently be used for the generation of chemicals, monomers, and polymers. Thermoset polymers that undergo hydrolysis and solvolysis under benign conditions typically contain degradable chemical groups: A variety of thermosets containing esters (24–26), acetals (27–29), carbonates (30), sulfur moieties (31–33), Schiff bases (34), and furans and maleimides linkages (35–37) have been reported. These crosslinked polymers were degraded in organic solvents or water (often accelerated with acids or bases) at low to moderate temperatures (typically <200°C and, in some cases, at room temperature). An important objective in the development of these materials is to create thermosets with desirable properties that are stable during their useful lifetimes in products and then can subsequently be degraded through hydrolysis or solvolysis once they become waste.

We first present the state-of-the-art knowledge in hydrolysis and solvolysis degradation mechanisms for both linear and thermoset polymers (**Figure 1**). We then discuss various factors that influence the hydrolysis or solvolysis rate in different classes of thermosets. Finally, we provide insight into recent studies that have demonstrated the preparation of degradable thermosets that also retain their useful physical properties, such as thermal and mechanical behavior, an essential feature for using these materials in applications. Throughout this review, we also highlight the incorporation of bio-based, plant-based, or biorenewable monomers into thermosets (21, 38–41), which often naturally contain degradable chemical groups (such as esters), and therefore not only address the end-of-life behavior of these polymers but also provide for their sustainable origin.

2. HYDROLYSIS AND SOLVOLYSIS DEGRADATION MECHANISMS IN LINEAR AND THERMOSET POLYMERS

2.1. Polyesters

The hydrolysis and solvolysis of linear polyesters have been explored extensively over the last 20 years. Linear polyesters undergo ester hydrolysis and solvolysis under benign conditions, at low

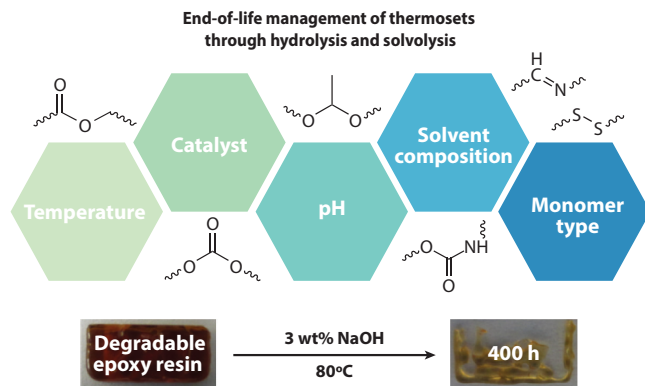


Figure 1

Mechanisms and factors affecting hydrolysis and solvolysis of thermosets, toward developing routes to their end-of-life disposal.

temperatures, and often assisted by the presence of catalysts (42–46). Throughout these decades of research, insight has been gained into degradation mechanisms. Von Burkersroda et al. (47) first introduced a theoretical model that includes a critical device dimension to predict the erosion mechanism of water-insoluble biodegradable polymer matrices, which depends on not only the hydrolysis or solvolysis reaction but also diffusion of water (or solvent) into the specimen. The authors proposed that all degradable polymers can undergo either surface or bulk erosion, depending on the degradation conditions, which induce disparate behaviors (**Figure 2**). In surface erosion, in which bond cleavage is faster than water diffusion into the specimen, hydrolysis/solvolysis occurs mainly in the region of the sample near the surface. By contrast, in bulk erosion, in which water diffusion into the specimen is more rapid than bond cleavage, hydrolysis/solvolysis occurs throughout the entire specimen simultaneously at a constant velocity throughout the erosion process. Polyesters such as polylactide and poly(lactide-*co*-glycolide) were shown to degrade differently in basic and acidic solutions (47) (**Figure 3a,b**). In basic solution, the degradation process commenced as soon as the samples were introduced to the solution, and the mass reduced linearly with degradation time (**Figure 3a**), characteristic of surface erosion. In acidic solution, the samples exhibited little mass loss for a period of time while the solution infiltrated them, after which an abrupt loss of sample mass was observed (**Figure 3b**), characteristic of bulk erosion.

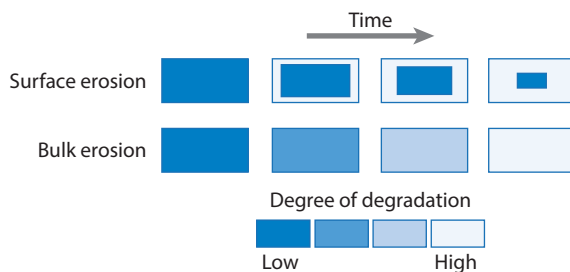


Figure 2

Schematic illustration of a polymer matrix that undergoes surface and bulk erosion. Reprinted from *Biomaterials*, 23, von Burkersroda F, Schedl L, Göpferich A. Why degradable polymers undergo surface erosion or bulk erosion, pp. 4221–31, copyright 2002, with permission from Elsevier (47).

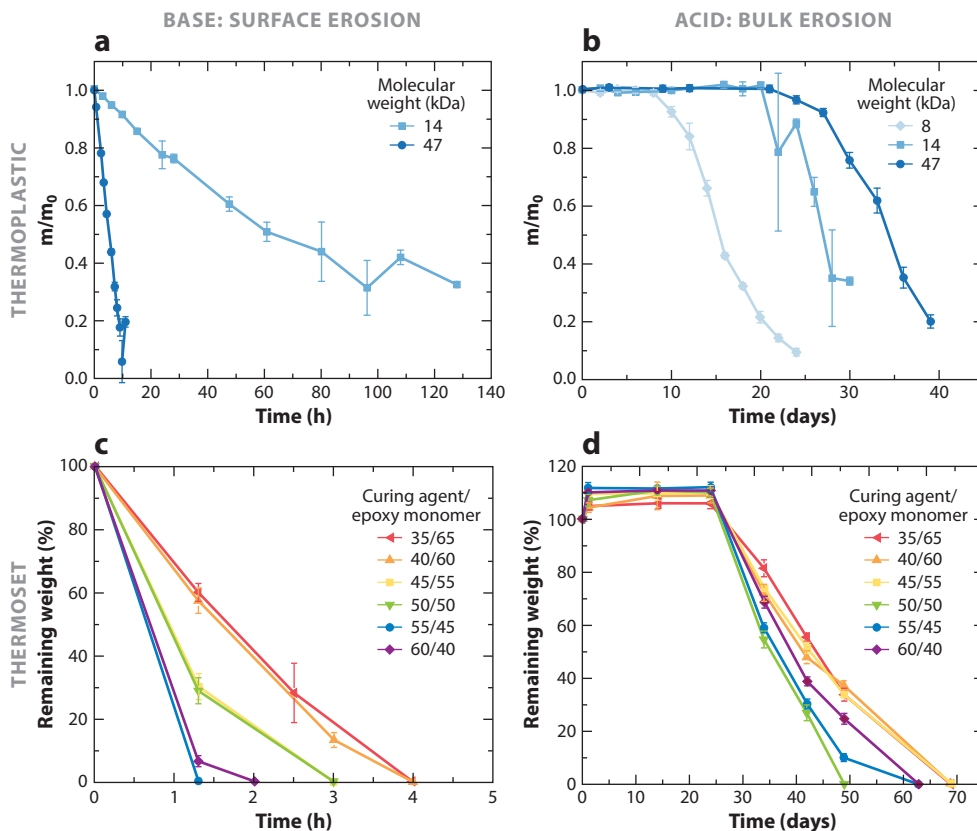


Figure 3

Thermoplastics and thermosets undergoing hydrolytic degradation, in basic and acidic solutions, exhibit features of surface and bulk erosion, respectively. Poly(lactide-co-glycolide) of varying molecular weights at (a) pH > 13 and (b) pH = 2 (reprinted from *Biomaterials*, 23, von Burkersroda F, Schedl L, Göpferich A. Why degradable polymers undergo surface erosion or bulk erosion, pp. 4221–31, copyright 2002, with permission from Elsevier) (47) (m_0 is the initial sample mass and m the sample mass at time t). Epoxy resin (trimethylolpropane triglycidyl cured with succinylated β -cyclodextrin at different ratios) degraded in (c) 1 M NaOH and (d) 1 M HCl (adapted with permission from Zhang Q, Molenda M, Reineke TM. 2016. *Macromolecules* 49:8397–406; copyright 2016 American Chemical Society) (65).

Once an ester bond is cleaved through hydrolysis or solvolysis, both a carboxylic group and a hydroxyl group are formed (**Figure 4**). The formed carboxylic acid groups auto-catalyze the subsequent cleavage of additional ester bonds (48). Various mathematical models have been established to describe the changes in the molecular weight distribution of linear polyesters (specifying changes in number-average molecular weight and weight-average molecular weight), based on assumptions regarding the reaction order (i.e., the role of both hydrogen/hydroxide ion and ester group concentrations); the presence of autocatalysis; relative rates of reaction and diffusion; diffusion of degradation products; role of material porosity; and erosion by diffusion of polymer, dissolution, or pore formation (44, 49–56) and have been reviewed previously (44, 49, 57–63).

More recently, the focus is turning to degradation mechanisms in crosslinked polyester thermosets. Two routes are possible for the preparation of thermosets containing degradable ester bonds. In some cases, monomers are used that contain ester functionality. Alternatively, the curing process itself may generate esters [as in the case of anhydride curing of an epoxy monomer (64)]. Zhang et al. (65) demonstrated that ester-containing epoxy resins exhibit very similar mass

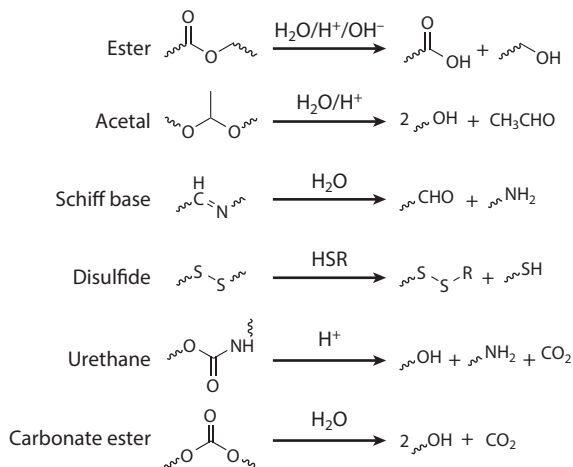


Figure 4

Degradable chemical groups incorporated into thermosets and their mechanisms of hydrolysis and solvolysis.

loss behavior to that of linear polyesters in basic and acidic solutions. In basic solution, the degradation behavior was consistent with surface erosion (**Figure 3c**), and in acidic solution, the degradation behavior was consistent with bulk erosion (**Figure 3d**). In the case of surface erosion in basic solutions, solid-state kinetic models were evaluated. Numerous possible solid-state kinetic models could be employed to understand surface erosion processes, such as n th-order kinetic models, diffusion models, and nucleation models (66, 67). Consistent with an earlier report by Kuang et al. (68), our group recently confirmed the use of the contracting volume solid-state kinetic model to describe the surface erosion of a variety of different types of epoxy resins in basic solutions (21) (**Figure 5**). The contracting volume model assumes that the rate-limiting process is the contraction of the sample surface toward its interior, and other processes, such as diffusion, nucleation, and even the reaction order, are secondary (69–71). We demonstrated that epoxy resins prepared with a conventional bisphenol A-derived monomer (the diglycidyl ether of bisphenol A, DGEBA, cured with an anhydride-curing agent), as well as those prepared from biorenewable replacements for DGEBA (derived from lignin, plant-based phenolic acids, and soybean oil), exhibited degradation behavior in basic solution consistent with the contracting volume model (21) (**Figure 5**). Interestingly, the biorenewable epoxy resins exhibited the fastest hydrolysis rates, attributed to the higher concentration of ester bonds throughout the epoxy networks as well as the enhanced hydrophilicity of the materials.

Various studies have identified the mechanism of ester hydrolysis or solvolysis in ester-containing epoxy resins. The cleavage of ester groups has been identified through Fourier-transform infrared spectroscopy (FTIR) analysis of the degrading specimen to show the disappearance of ester groups, or alternatively through nuclear magnetic resonance (NMR) analysis of degradation products. In one study, DGEBA copolymerized with γ -butyrolactone) was hydrolyzed in 1 M ethanolic potassium hydroxide (KOH) at 80°C for 24 h, after which characteristic peaks associated with the ester bond disappeared (72). In another study, the glycolysis of an anhydride-cured DGEBA epoxy resin in the presence of titanium n -butoxide/diethylene glycol at 245°C, over a time period of 1.5–8 h, was confirmed through FTIR, and the alcoholysis products were identified as esterdiols (73). The depolymerization of an anhydride-cured DGEBA epoxy resin in a 2-ethylhexanol solution containing triazabicyclodecene catalyst at 160–180°C,

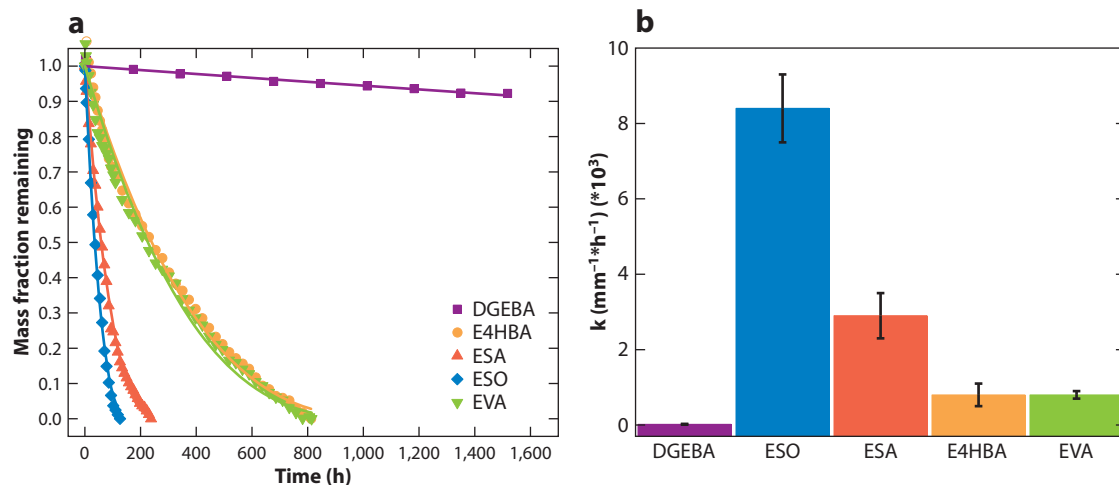


Figure 5

(a) Contract volume model fitting of diglycidyl ether of bisphenol A (DGEBA) and bio-based epoxy resins [epoxidized vanillic acid (EVA) derived from lignin, epoxidized 4-hydroxybenzoic acid (E4HBA) and epoxidized salicylic acid (ESA) derived from bio-based phenolic acids, epoxidized soybean oil (ESO)]. (b) Degradation rate constants of epoxy resins extracted from fitting the contracting volume model to the data. Adapted from Reference 21 with permission from The Royal Society of Chemistry.

which occurred between 2–5 h, was characterized with FTIR and ^1H -NMR (68, 74). The presence of both free and end-terminated ethylene glycol directly confirmed the ester solvolysis mechanism. The degradation of an epoxy resin composed of a eugenol-based epoxy monomer cured with succinic anhydride, which degraded in ethanol within 7 h at 160°C without catalyst, was confirmed through ^1H -NMR (24). Anhydride-cured epoxy resins degraded within 5 h in aqueous phosphotungstic acid solution at 200°C , and the disappearance of ester groups in the specimen, along with the appearance of carboxylic acid groups and primary and secondary hydroxyl groups, was characterized through FTIR and ^{13}C -NMR (75).

Degradation products may also be analyzed through mass spectrometry (MS). A citric acid-based epoxy resin cured with an anhydride was fully degraded within 2 h at 90°C in hydrogen peroxide (H_2O_2)/dimethylformamide (DMF), in which ester cleavage was confirmed through gas chromatography–MS (GC-MS) (26). Along with cleavage of ester groups, hydroxyl oxidation and amino oxidation occurred owing to the strong oxidation capabilities of H_2O_2 . Similarly, our group also identified the hydrolysis of ester bonds in various biorenewable epoxy resins [derived from vanillic acid, phenolic acids, and epoxidized soybean oil (ESO)] through characterization of degradation products using MS (21). Finally, the physical properties of epoxy resins change during the degradation process. DGEBA epoxy resins containing dendritic poly(ϵ -caprolactone) stars exhibited enhanced reworkability, with significant loss of glass transition temperature (T_g , an indication of reduced crosslink density) after exposure to a basic solution (76).

2.2. Polyacetals

Degradable polymers, both linear polymers and thermosets, have also been created through the incorporation of acetal groups within the materials. The acetal group also can be hydrolyzed, particularly in acidic conditions (Figure 4). A recent review paper discussed new routes to creating thermoplastic polyacetals, including those from sustainable sources (77).

Numerous studies have focused on degradation mechanisms in linear polyacetals (78, 79). The acid-catalyzed acetal hydrolysis proceeds in three steps: (a) the formation of an oxocarbenium ion, (b) the formation of hemiacetals, and (c) the formation of an aldehyde and diol. Among these three steps, the stability of oxocarbenium ion intermediates can largely determine the degradation rate, and the hydrolysis in the following two steps usually occurs more rapidly (80). Investigation of the degradation behavior of a series of small acetal molecules led to the following conclusions: (a) Increased steric strain can slow down the degradation rate, and *trans* derivatives degrade faster than *cis* derivatives; (b) when the pH is in the range of 1–6, hydronium ion- and water-catalyzed hydrolysis of the hemiacetal intermediate is the rate-determining step for degradation; and (c) when pH > 6, attack of H₂O on the oxocarbenium ion intermediate largely limits the degradation rate (81).

Degradation mechanisms in thermosets containing acetal functionality have also been explored. An acetal-containing epoxy resin, degraded within 12 h in 0.1 M hydrochloric acid in tetrahydrofuran (THF)/water at room temperature, was characterized through ¹H-NMR and gel permeation chromatography (measuring molecular weight of degradation products) to identify the acetal hydrolysis mechanism and quantify degradation products (27). A furfural-based cycloaliphatic epoxy resin, cured with an anhydride-curing agent, was fully degraded within several hours when exposed to an acidic aqueous solution also containing THF, and the degradation products were analyzed with FTIR and GC-MS (82). Vanillin-based epoxy resins containing spiro diacetal degraded within several minutes in an organic solvent containing hydrochloric acid and real-time ¹H-NMR confirmed that the spiro diacetal structure was cleaved to a monoacetal and reached an equilibrium concentration before further degrading (28). The spiro diacetal epoxy resin showed good resistance to water and basic solutions, with no significant decrease in T_g. A dicyclo diacetal-containing epoxy resin was degraded within 40 min in 1 M hydrogen chloride (HCl) acetone/water, and cleavage of acetal groups and the subsequent formation of formyl groups were also quantified with real-time ¹H-NMR (83).

2.3. Polyurethanes and Polyhexahydrotriazines

Owing to the stabilization effects induced by their dual amide structure, polyureas usually have good stability against hydrolysis and solvolysis. Urea bonds were destabilized by the introduction of bulky *tert*-butyl substituents, which promoted bond dissociation and hydrolytic degradation (84) (**Figure 4**). Crosslinked polymers based on 1,1-*tert*-butylethylurea degraded within 4 days in phosphate-buffered saline solution, and the cleavage of urea bonds was identified through ¹H-NMR. The degradation kinetics in crosslinked polyureas were tuned by modifying the steric hindrance of the monomer structure, in which bulkier structures led to faster degradation (85, 86). Polyhydroxyurethanes derived from six-membered cyclic carbonates and amines were fully degraded in tetraethylene glycol at 180°C within 4 h (87). The degradation mechanism was identified as transcarbamoylation of carbamates to isocyanates and alcohols through FTIR and density functional theory. The transcarbamoylation reaction was also proven to engender shape memory performance to polyurethane networks (17, 88).

2.4. Sulfur-Containing Polymers

With elemental sulfur produced as a large-scale industrial waste, increased emphasis is being placed on the use of sulfur-containing bonds to make degradable networks (**Figure 4**). Sastri & Tesoro (89, 90) first used disulfide linkages to recycle disulfide-containing epoxy resins through oxidation reactions, quantified using FTIR. The disulfide-containing epoxy resins were degraded by reducing the disulfide linkages to thiols. Sulfide-containing epoxy resins were also degraded in

the presence of 2-mercaptoethanol (91) and in 2-mercaptoethanol/DMF solutions, as well as in the presence of a thiol, owing to the thiol–disulfide exchange reaction, which disrupt crosslinks (92).

2.5. Polyimines

The imine moiety, which is also known as a Schiff base (**Figure 4**), is often used to develop dynamic structures that can be degraded under mild conditions. Schiff base covalent adaptable networks could be degraded within 12 h in 1 M HCl, quantified through real-time ^1H -NMR (93). The imine group was cleaved to amine and aldehyde groups. Harada et al. (34) used 4,4'-diaminodiphenylmethane as a curing agent to create a Schiff base structure in an epoxy resin that could be degraded in acidic solution after 28 days. The degradation products were dissolved in the HCl solution ($\text{pH} = 0.3$), and cleavage of the Schiff base bond was identified through FTIR and gel permeation chromatography. DGEBA cured with a dynamic hardener (vanillin and diamine), to create a Schiff base structure, underwent hydrolysis in acidic solution to release aldehydes and amines as degradation products, characterized through FTIR (94). An epoxy resin bearing aromatic imine bonds was hydrolyzed in organic and aqueous solutions under mild conditions, characterized through ^{13}C -NMR. Subsequently, the degradation by-products were extracted from the solution and could be cured to form a new epoxy resin, demonstrating the complete recycling of the material (95).

3. FACTORS AFFECTING HYDROLYSIS AND SOLVOLYSIS RATES IN THERMOSET POLYMERS

3.1. Effect of pH

Solution pH plays an important role in controlling degradation mechanisms and rates. Acid or base may act as a catalyst to promote hydrolysis or solvolysis reactions. Epoxy resins bearing aromatic imine bonds exhibited accelerated degradation rates in a variety of solvents with the addition of 0.17 M HCl (95). The degradation time of an acetal-containing vanillin-based epoxy resin decreased from 19 to 10 min as the concentration of HCl was increased from 0.1 to 0.5 M and then further decreased to 8 min at 1.0 M HCl (28) (**Figure 6a**). The degradation rate increased for ester-containing isosorbide-based and sucrose soyate-based epoxy resins when the sodium hydroxide (NaOH) concentration increased from 0.1 to 10 M (96). These studies demonstrate the acceleration of degradation rates in the presence of acid or base; however, increasing the acid or base concentration can also have the effect of reducing the solubility of the degradation products. In this case, the degradation rate may decrease, as shown in the case of a carbon fiber–reinforced epoxy resin containing imine groups for which the degradation time increased from 340 to 718 min when the HCl concentration increased from 0.1 to 1.0 M (93).

As discussed in Section 2, the degradation mechanisms in basic and acidic solutions are quite different from one another: Degradation proceeds as surface erosion in basic solution and bulk erosion in acidic solution for ester-containing polymers. Additionally, the degradation rates can be quite different under basic and acidic conditions. The degradation of ester-containing sugar-derived epoxy resins completed within hours in 1 M NaOH and within 50–70 days in 1 M HCl (65). Similarly, ester-containing epoxy resins derived from trehalose, ESO, and β -cyclodextrin could be degraded within several hours in basic solution, whereas no degradation occurred in neutral water or acidic solution (97). Ester-containing isosorbide and sucrose soyate-based epoxy resins degraded within 12 min in 1 M NaOH and took 48 h to degrade in 1 M HCl (25). An isosorbide diferulate-based epoxy resin degraded fully within 40 h in an NaOH solution and exhibited relatively slow degradation in HCl (98). The acid strength can also play a role in determining

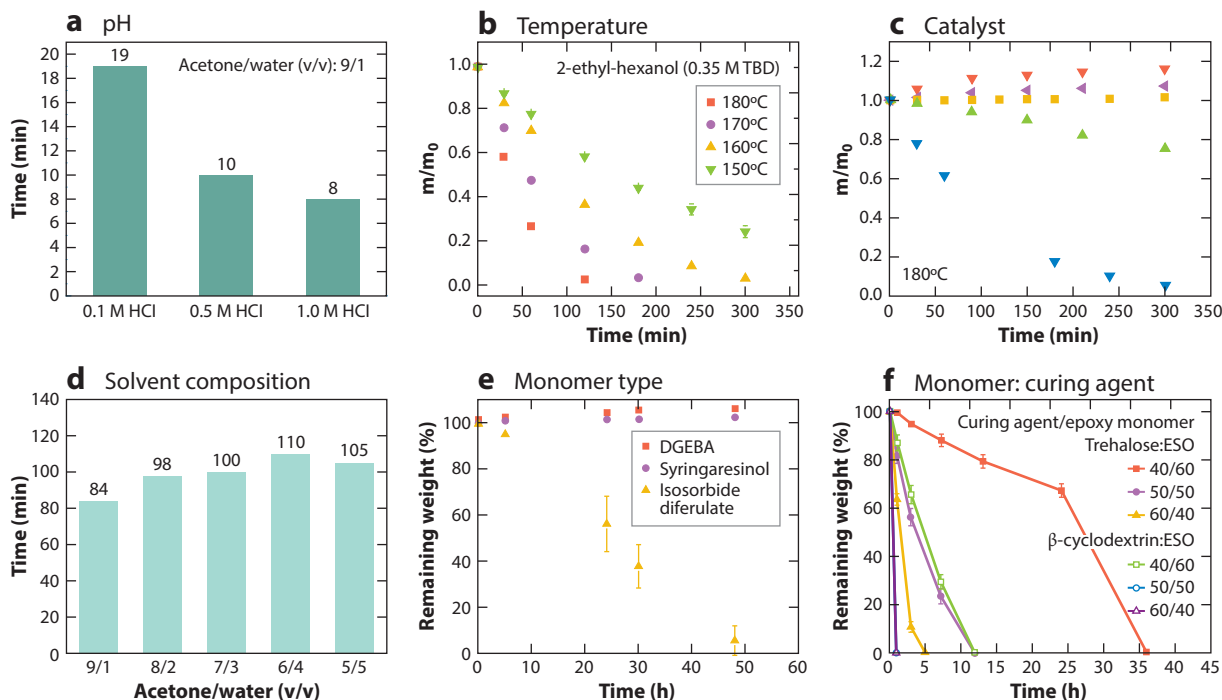


Figure 6

Different factors influence degradation rates. (a) Effect of pH: degradation of acetal-containing epoxy resins in acidic aqueous solutions (adapted from Reference 28 with permission from The Royal Society of Chemistry). (b) Effect of temperature: degradation of a diglycidyl ether of bisphenol A (DGEBA)-based epoxy resin in 0.35 M triazabicyclodecene (TBD) in 2-ethyl-hexanol (reproduced with permission from Kuang X, Shi Q, Zhou Y, Zhao Z, Wang T, Qi HJ. 2018. *RSC Adv.* 8:1493–502. Published by the Royal Society of Chemistry) (68). (c) Effect of catalyst: degradation of a DGEBA-based epoxy resin at 180°C in the presence of 0.35 M triphenylphosphine (red triangles), zinc acetylacetonate (purple triangles), NaOH (green triangles), and TBD (blue triangles), and without catalyst (gold squares) (reproduced with permission from Kuang X, Shi Q, Zhou Y, Zhao Z, Wang T, Qi HJ. 2018. *RSC Adv.* 8:1493–502. Published by the Royal Society of Chemistry) (68). (d) Effect of solvent composition: degradation of a diacetal-containing epoxy resin cured with diaminodiphenyl sulfone in acetone/water solutions (adapted from *Eur. Polym. J.*, 117, Yuan W, Ma S, Wang S, Li Q, Wang B. Synthesis of fully bio-based diepoxy monomer with dicyclo diacetal for high-performance, readily degradable thermosets, pp. 200–7; copyright 2019, with permission from Elsevier) (83). (e) Effect of monomer type: degradation of epoxy resins prepared from various epoxy monomers (DGEBA, syringaresinol, and isosorbide diferulate) (98; copyright 2017 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim). (f) Effect of monomer-to-curing agent ratio: degradation of ESO-based epoxy resins cured with trehalose (closed symbols) and β -cyclodextrin (open symbols) (adapted with permission from Zhang Q, Phillips HR, Purchel A, Hexum JK, Reineke TM. 2018. *ACS Sustain. Chem. Eng.* 6:14967–78; copyright 2018 American Chemical Society) (97).

degradation rate: A diepoxy monomer containing bicyclo diacetal could be fully degraded within 40 min in 1 M HCl, and the degradation time increased to 70 min in 1 M sulfuric acid (H_2SO_4) and 480 min in phosphoric acid (H_3PO_4). In the hydrolysis of acetals, both the oxocarbenium formation and hemiacetal cleavage require sufficient H^+ concentration; however, the medium-strong acid H_3PO_4 could protonate only partially compared with fully protonated HCl and H_2SO_4 at the same solution concentrations (83).

3.2. Effect of Temperature

Degradation temperature is an important factor that will determine energy costs associated with disposing of thermosets through hydrolysis or solvolysis. There is much demand for the ability

to degrade thermoset polymers at relatively low temperatures. Citric acid-based epoxy resins containing esters cured with an anhydride-curing agent lost 46.4% of their mass at 50°C in H₂O₂/DMF, and this was increased to a mass loss of 99.9% at 90°C (26). Similarly, the hydrolysis rate of a vanillin-based epoxy resin containing acetal groups increased by a factor of 10 in an acetone/water solution when the temperature was increased from room temperature to 50°C (28). An epoxy thermoset bearing aromatic imine bonds degraded more rapidly at 65°C relative to 25°C (95). An anhydride-cured DGEBA-based epoxy resin, depolymerized by selective ester bond cleavage in a triazabicyclodecene (TBD)/alcohol solution, showed a factor-of-5 increase in degradation rate as the temperature was raised from 150°C to 180°C (68) (**Figure 6b**).

Bio-based epoxy resins that contain cleavable bonds (esters, acetals, imines) typically require significantly lower hydrolysis or solvolysis temperatures than those of commercial DGEBA (or other bisphenol monomer) epoxy resins. Commercial epoxy resins usually require temperatures of approximately 300°C, combined with high pressure, for hydrolysis or solvolysis to proceed (99, 100). By contrast, the literature provides many examples of bio-based epoxy resins that can degrade at room temperature or slightly elevated temperatures. Imine-containing bio-based epoxy resins underwent hydrolysis at room temperature in 0.1 M HCl and fully degraded within 4 h (93). Similarly, a soybean oil-based epoxy resin degraded fully at room temperature in 1 M NaOH within 1 h (97). An acetal-containing bio-based epoxy resin degraded within 1 h at 50°C in 0.1 M HCl in DMF (83).

3.3. Effect of Catalyst

Developing an efficient catalyst or optimizing the appropriate amount of catalyst is crucial to identifying degradation protocols that operate at lower temperatures, and thus lower energy cost, as well as use more benign conditions (reducing acid or base concentration and avoiding toxic solvents). A citric acid-based epoxy resin containing ester groups, cured with an anhydride-curing agent, could be fully degraded within 2 h at 90°C in DMF when H₂O₂ was present, which produces free radicals and promotes decomposition (26). The presence of CO₂ in water and an acetone/water (80/20) solution enhanced resin degradation under conditions of high temperature and high pressure, where CO₂ acted as a catalyst in the presence of the phenol produced by the resin degradation (101). The degradation of carbon fiber-reinforced amine-cured epoxy resins occurred in water at supercritical or near-critical conditions: Only 79.3 wt% of the sample mass was degraded in supercritical water conditions, and this increased to 95.3 wt% with the addition of KOH as an alkali catalyst (102). In the degradation of anhydride-cured DGEBA-based epoxy resins, TBD acted as a strongly basic catalyst and promoted cleavage of ester groups (68, 74) (**Figure 6c**). A weak Lewis acid (zinc chloride, magnesium chloride, and aluminum chloride) promoted degradation of a carbon fiber-reinforced amine-cured epoxy resin in acetone/water by reducing the reaction activation energy (99). The addition of KOH reduced the initial decomposition temperature of an amine-cured epoxy resin by 10–30°C and promoted the formation of complex alcohols (103).

3.4. Effect of Solvent Composition

Changing the solvent system has a large impact on degradation. An important effect of the choice of solvent is the solubility of degradation products; degradation rates are inhibited when degradation products are insoluble. Vanillin-based epoxy resins were degraded in a variety of solvents containing HCl; degradation proceeded much more slowly in ethanol as compared with that in methanol, acetone, THF, and DMF, attributed to the significantly worse solubility of the

degradation products in ethanol (93). Similarly, in another study, a vanillin-based epoxy resin containing an imine group showed very slow hydrolysis rates in toluene and benzene owing to the poor solubility of degradation products in these two solvents (95). In the degradation of an amine-cured commercial epoxy resin, it was discovered that the optimal degradation conditions occurred in acetone/water (80/20) for which the solubility of degradation products was greatest (101). Similarly, in an acetal-containing epoxy resin, acetone/water (90/10) was the most effective solvent (83) (**Figure 6d**).

The affinity of thermoset polymers to the solvents also plays an important role in controlling solvolysis rates, as increased wettability of the thermoset surface to the solvent enhances the degradation rate. An imine-containing epoxy resin showed a decrease in solvolysis rate as the polarity of the solvent and affinity of thermoset to the solvent decreased (water > dimethyl sulfoxide > DMF > ethanol > THF > benzene) (95). An acetal-containing epoxy resin degraded more rapidly when the solvent polarity increased (83). Ethylene glycol monobutyl ether degraded an anhydride-cured DGEBA epoxy resin faster than other alcohols owing to the higher affinity of the solvent to the epoxy resin (68). Finally, adding THF or acetone to water improved the hydrolysis rate of an amine-cured commercial epoxy resin (104).

3.5. Effect of Type and Concentration of Monomer and Curing Agent

Varying the monomer type can have a larger impact on the degradation rate. In comparisons of various bio-based epoxy resins derived from vanillic acid, bio-based phenolic acids, and soybean oil, increasing the number of esters per epoxy monomer and increasing the epoxy resin hydrophilicity were found to increase the degradation rate (21). An isosorbide diferulate-based epoxy resin that contained esters could be fully degraded in HCl solution within 40 h, whereas a syringaresinol-based epoxy resin that did not contain any cleavable groups degraded much more slowly (98) (**Figure 6e**). Choice of curing agent may also influence degradation rate: The degradation rate increased for sulfide-containing epoxy resins in 2-mercaptoethanol when the concentration of disulfide bonds in the curing agent increased (91). When ESO was cured with trehalose, a more hydrophilic resin was produced as compared with curing with β -cyclodextrin, and the trehalose-cured resin degraded more rapidly (97).

Varying the ratio of epoxy monomer to curing agent can also tune the degradation rate of an epoxy resin. In sucrose soyate-based epoxy resins, as the ratio of the monomer to the curing agent increased, the degradation rate decreased (96). ESO-based epoxy resins cured with trehalose and β -cyclodextrin degraded more rapidly when the ESO content was reduced, owing to the high ester content in the curing agents (97) (**Figure 6f**).

4. DEGRADABLE THERMOSETS THAT MAINTAIN THERMAL, MECHANICAL, AND ADHESION PROPERTIES

The previous sections illustrate how the incorporation of appropriate functional groups within the thermoset network is a viable route to the preparation of thermosets that degrade under relatively benign conditions. A great challenge is to develop strategies for creating degradable thermoset polymers without compromising their performance, and which exhibit desirable thermal, mechanical, adhesion, and other properties. In this section, we present a few notable examples of thermoset polymers containing hydrolytically or solvolytically cleavable bonds that can maintain needed properties.

Numerous studies have shown how ester-containing thermosets can be both hydrolytically and solvolytically degradable and also exhibit excellent physical properties. Our group previously

Table 1 Thermal and mechanical properties of degradable epoxy resins (ESA and E4HBA), as compared to that of a DGEBA epoxy resin^a

Monomer	T _g (°)	Tensile strength (GPa)	Modulus (GPa)	Strain at break (%)	Tensile toughness (MPa)
ESA	131 ± 3	85 ± 6	3.0 ± 0.3	5 ± 1	3 ± 1
E4HBA	136 ± 5	90 ± 2	2.9 ± 0.2	8 ± 1	5 ± 1
DGEBA	140.1 ± 0.4	80 ± 2	2.5 ± 0.2	8 ± 2	5 ± 1

^aDegradation rate constants of these materials are provided in **Figure 5** (adapted with permission from Reference 41; copyright 2016 American Chemical Society).

Abbreviations: DGEBA, diglycidyl ether of bisphenol A; E4HBA, epoxidized 4-hydroxybenzoic acid; ESA, epoxidized salicylic acid; T_g, glass transition temperature.

synthesized degradable bio-based epoxy resins (derived from plant-based phenolic acids) in which the chemical structure of the bio-based monomer mimicked that of DGEBA. The resulting epoxy resins exhibited mechanical and thermal properties comparable to those of DGEBA (21, 41) (**Table 1**). ESO is generally a more flexible epoxy monomer, and the resulting epoxy resins exhibit lower crosslink density, T_g, and modulus (105). To overcome this limitation, ESO was cured with a succinylated β-cyclodextrin hardener and showed significantly improved properties (tensile strength increased from 10 to 64 MPa). Furthermore, the lap shear adhesion was in the range of 2.1–3.6 kpsi, which is close to that of commercial epoxy adhesives (2.9–3.8 kpsi) (65). In another example, a hyperbranched aromatic polyester with terminal hydroxyl groups (HBPCL) was incorporated into a DGEBA-based epoxy resin. The addition of HBPCL accelerated the degradation rate and enhanced the Young's modulus (increased from 2.5 to 3.1 GPa) (76).

Acetal-containing thermosets also show much promise for the creation of degradable thermosets that maintain physical properties. Degradable spiro diacetal-containing epoxy resins exhibited comparable thermal and mechanical properties (T_g = 164°C, tensile strength = 85 MPa, Young's modulus = 3.1 GPa) and comparable carbon fiber composite mechanical properties (tensile strength = 731 MPa, modulus = 40 GPa) to that of conventional epoxy resins (28) (**Figure 7**). Similarly, a dicyclo diacetal-containing epoxy resin exhibited excellent performance (tensile modulus = 3.3 GPa, tensile strength = 72 MPa, T_g = 160°C) (83). The high performance of these epoxy resins was due to the rigidity of the diacetal structures. Acetal-linked furfural-based epoxy resins also exhibited desirable properties (T_g = 186°C, storage modulus = 2.51 GPa in the glassy region and 16.5 MPa in the rubbery region) (82). Cresol novolak-type phenolic resins containing degradable acetal linkages exhibited high tensile properties (tensile strength = 65.44 MPa, Young's modulus = 2.96 GPa) (29).

The presence of imine (Schiff base) bonds provides a route to create degradable epoxy resins that are also reprocessable and exhibit excellent mechanical and thermal performance. A Schiff base-containing epoxy resin showed desirable tensile strength (85 MPa) (94), which is similar to that of epoxy resins composed of ester or disulfide bonds. In another study, the T_g of a Schiff base-containing epoxy resin was as high as 210°C (34). Finally, Schiff base covalent adaptable epoxy networks showed high performance (Young's modulus = 2.11 GPa, tensile strength = 81 MPa, T_g = 172°C) (93). Owing to the existence of Schiff base bonds, these epoxy resins can be easily reprocessed under acidic conditions and achieve excellent retention properties. In one study, not only did the material properties not suffer during the reprocessing cycle, but slight improvements were observed: The Young's modulus increased from 2.11 to 2.28 GPa, the tensile strength was maintained at 81 MPa, and the T_g increased from 172°C to 175°C (93). Epoxy resins bearing aromatic imine bonds maintained tensile strength (46 MPa) through reprocessing (95).

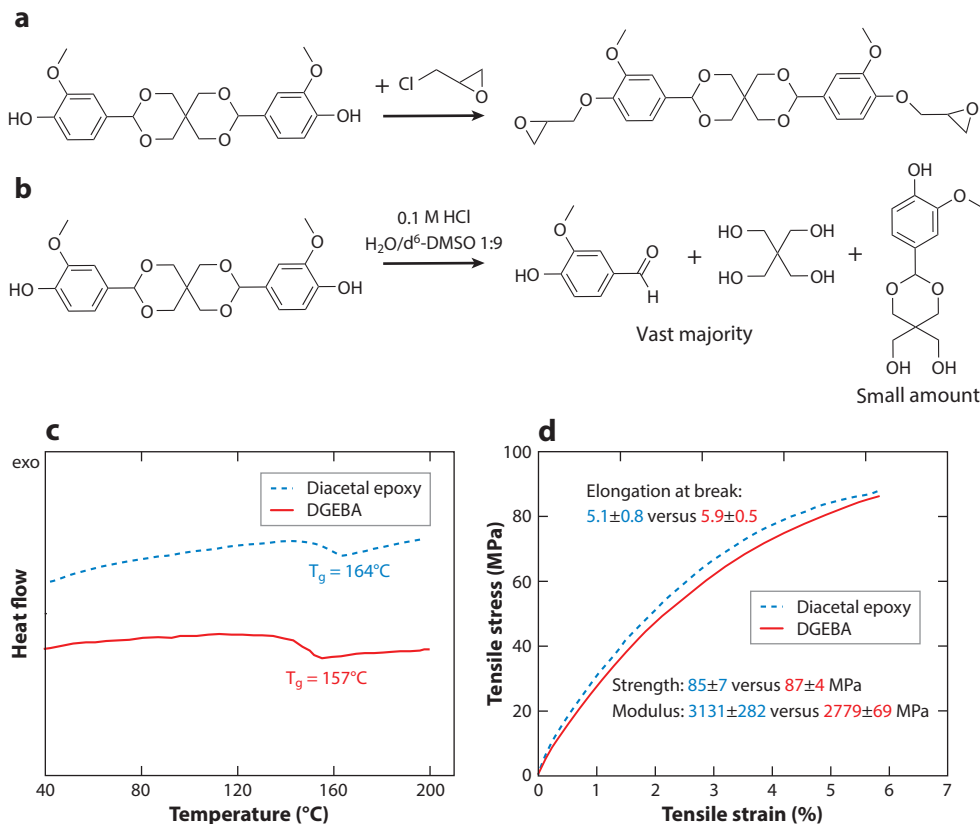


Figure 7

(a) Synthesis of diacetal-containing epoxy monomer. (b) Degradation mechanism of diacetal-containing epoxy monomer in acidic solution. Degradable epoxy resins prepared from the diacetal monomer exhibit (c) thermal and (d) mechanical properties similar to a diglycidyl ether of bisphenol A (DGEBA)-based epoxy resin. Reproduced from Reference 28 with permission from The Royal Society of Chemistry.

Hexahydro-*s*-triazine (HT)-containing thermosets showed good thermal and mechanical properties, as well as degradability under strongly acidic conditions (106). An epoxy resin with HT structure exhibited desirable physical properties (Young's modulus = 2 GPa, tensile strength = 80 MPa, T_g = 151°C) (107). Thermosets containing urethane bonds also exhibited good properties (Young's modulus = 2.2 GPa, tensile strength = 72 MPa, and T_g = 54°C) (87).

The addition of disulfide groups into the thermoset can promote degradation owing to associative crosslink exchange, while not limiting physical properties and performance. Johnson et al. (91) evaluated the influence of different contents of disulfide bonds in the system and found that addition of disulfide bonds did not affect the T_g (approximately 140°C) and storage modulus (approximately 2.5 GPa). Ruiz de Luzuriaga et al. (92) also reported the high performance of a disulfide bond-containing epoxy resin (tensile strength = 88 MPa, T_g = 130°C).

5. SUMMARY AND OUTLOOK

Thermoset polymers that contain cleavable groups degrade through hydrolysis and solvolysis under relatively benign conditions, not unlike their thermoplastic counterparts. Degradable

thermosets have been developed with a variety of chemical functionalities: esters, acetals, imines, carbonates, sulfur moieties, furans, maleimides, and others. Thermoset degradation mechanisms in basic and acidic solutions follow those of surface and bulk erosion, respectively. A variety of methods can be applied for tuning the thermoset degradation rate: varying pH and temperature; adding catalysts; using different solvent compositions; and varying the choice of monomer, curing agent, and monomer-to-curing agent ratio. This body of literature demonstrates that incorporating the appropriate chemical functionalities into thermoset polymers is a viable route to the facile degradation of these materials under the right conditions. To be useful for the end-of-life management of thermoset waste, however, further issues must be addressed. Ideally, the thermosets would still exhibit excellent stability in environmental conditions, preventing degradation during the product lifetime and avoiding deterioration of properties, and this aspect has not been investigated extensively. It may be possible to create a thermoset that can degrade under the right conditions (elevated temperature, variations in pH) while still maintaining its performance under ambient conditions, but the practicality of this approach remains an open debate. The design of thermosets for their end of life, whether through the creation of degradable or reprocessable materials, must be considered in the early stages of material design. To promote widespread use of degradable thermosets in applications, further investigations are required into their physical properties to ensure they are viable replacements for the variety of thermosets used commercially. Relevant properties go beyond commonly characterized thermal and mechanical behavior and include physical properties relevant to product manufacturing, such as viscosity changes during curing, adhesion to fillers, and behavior during composite manufacturing.

To address the full life cycle and overall sustainability of degradable thermosets, numerous factors must be considered. The incorporation of bio-based monomers into thermosets to impart degradability addresses not only a material's end of life but also its origin. Whether bio-based thermosets can be adopted in applications will depend on other factors, such as resource availability, maintenance of desired physical properties, and cost. The impact of degradation products on the environment, as well as on human health, must also be considered (especially if degradation can proceed within waterways). The design of degradable thermosets that degrade into benign by-products is an important objective. Optimally, degradation products can be used as feedstocks for the development of new polymers, completing the material life cycle, and this is an area for which new research can make a significant impact.

SUMMARY POINTS

1. Thermosets that contain degradable groups (e.g., ester, acetal, and imine) undergo hydrolytic and solvolytic degradation.
2. Thermosets containing ester groups undergo surface erosion in basic solutions and bulk erosion in acidic solutions, in analogy to the behavior of linear polyesters.
3. Solid-state kinetic models describe the degradation behavior of thermosets undergoing surface erosion.
4. The hydrolysis or solvolysis rates of thermosets are influenced by a variety of factors: pH, temperature, monomer type, curing agent type, monomer-to-curing agent ratio, presence of catalyst, and solvent composition.
5. Hydrolytically or solvolytically degradable thermosets can be designed to exhibit needed physical properties, such as thermal, mechanical, and adhesion behavior.

6. In many cases, bio-based thermosets contain needed functionality (such as esters) to promote degradation through hydrolysis or solvolysis.
7. The relatively benign conditions required for degradation of thermosets that contain appropriate chemical groups (e.g., esters and acetals) may provide a route for disposal of thermosets at the end of their useful lifetime.

DISCLOSURE STATEMENT

The authors are not aware of any affiliations, memberships, funding, or financial holdings that might be perceived as affecting the objectivity of this review.

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